PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Co-design approaches involving older adults in the development of electronic healthcare tools: a systematic review
AUTHORS	Cole, Amy; Adapa, Karthik; Khasawneh, Amro; Richardson, Daniel; Mazur, Lukasz

VERSION 1 – REVIEW

REVIEWER	de Mul, Marleen Erasmus Universiteit Rotterdam Erasmus School of Health Policy and Management
REVIEW RETURNED	19-Jan-2022

GENERAL COMMENTS	Understanding how co-design works in practice for different sub-groups is a relevant topic. The researchers have followed a well-documented, thorough process to select papers for their review, with the aim of providing further insight into co-design. I appreciate that the authors also assessed the quality of the included papers. However, I am puzzled by the purpose of this particular review, and how the result of this study adds additional insights to Fischer's much-cited review. The aim is promising, but for me the analysis does not yet meet my expectations. This may have to do with the way the results are presented, and the choice of counting over narrative description. Counting how many different frameworks there are, or what the average number of phases is in co-design projects with older adults, is -in itself - not very interesting to me. It does not help me to better understand the specifics, the challenges, and the work required to create value from co-design with older people. I advise the authors to analyse their selected articles in other ways and also to present the data they have found in a different way. This will make the study more relevant to the readership of BMJ Open, who are not designers and therefore lack the ability to pick up on the nuances of co-design terminology. Suggestion 1. At present, the emphasis in the first part of the results is on distinguishing the methodological terms used by the authors of the original studies. Without clarification of what these terms mean and, for example, ordering these terms into classes, it remains a random (alphabetical) list and there is, for example, no clear substantive difference between table 5 and tables 6 and 7. It is unclear what a reader should do with this information. Table 9 is informative, but cross-table 10 not so much. What do they show? I recommend the current Table 4 be expanded, to include information per study on the co-design approach chosen, before the aggregate data is presented in later tables. If the co-design

	methodology is and will remain a central theme in the paper, I am interested not only in the aggregated data, but also in the original data. Suggestion 2. You could analyse the selected papers for working mechanisms, how the authors wrote about the involvement of older people in different phases of their project. Was it different from other target groups? How did the researchers specifically take into account the characteristics of older people such as lower health literacy, anxiety for technology, endurance… when deciding where to involve older adults and where not, or when deciding to use tool A rather than tool B. This requires a different analysis of the text, going beyond what can be counted and averaged. Suggestion 3. Critically review the classifications: a. The study by Herrmann et al. is used to classify the needs of older adults addressed by EHTs. I wonder if it is relevant to use an additional classification for the EHT included in the evaluation (on top of the WHO classification), as this is not the main subject of the evaluation. Another reason why combining these two frameworks is less valuable is that both frameworks relate to care innovation in general, not specifically to older people. In my opinion it would be sufficient to use either WHO or Herrmann and not both. We know that there are many more studies on EHT, some even specifically for older people. However, they were not included in this review because no co-design methodology was described. The type of EHT is not a relevant result, but a characteristic of the selected papers. Therefore, I would add this information to the first section of the results or an appendix, not to the last section (implying that it is the climax of the study). The cross-classification in Table 15 is not relevant for the purpose of this article. b. I like the idea of going deeper into how, when and how intensively the older adults are involved. However, I think it would be of value to first apply Vaughn's model to the different co-design methodologies (for example, in human-centered design the role of the user in development is different than in co-creation). In Sanders & Stappers' book, there is also a matrix that positions different design methodologies in terms of user involvement. This would make the decision on which level to place a particular paper more transparent and replicable. Table 12 and the text on page 10, lines 41-49 are not clear. Other comments: Looking at the search string, I wondered if papers on personal health records and apps were also retrieved. These also seem like important tools to inform healthcare decisions. Minor: The section on strengths and limitations seems difficult to read, due to incomplete sentences and lack of signal words Add a column with the reference number in Table 4. The headings could also be more self-explanatory as this is not a review of a clinical study. "No stakeholders" makes more sense than "sample size". References 47 and 51 are the same.

REVIEWER	Otero Varela, Lucía

	University of Calgary, Community Health Sciences
REVIEW RETURNED	31-Jan-2022

GENERAL COMMENTS	This is a systematic review on "Co-design approaches involving older adults in the development of electronic healthcare tools", submitted to BMJ Open by Cole et al. The authors have thoroughly followed a rigorous methodology to synthesize the current state of the literature on a novel area that involves a growing population (adults aged 60 and older). Adapting electronic healthcare tools to meet their needs is imperative, and thus, the topic of this review is relevant and interesting to the BMJ Open audience, which makes this manuscript a worthwhile piece of translational work. Nonetheless, although the complexity of this topic should be acknowledged, there is potential for improvement, especially in terms of reporting. The authors summarize the manuscript appropriately in the abstract, and provide a comprehensive introduction to contextualize the focus of the review. However, the methods must be better described, placing focus in the outcomes and data synthesis, and the results need to be clearly presented. Special attention should be given to the tables (too numerous): firstly, there are too many, some should be combined or deleted; secondly, the results table is missing, which is key to understand the findings from this review; and thirdly, there are tables from which no inferences are drawn, thus, the authors should reflect on whether these were necessary or should be deleted. For detailed recommendations, please see my comments below, categorized by manuscript section. Introduction - Page 3, line 33. The opening sentence of the introduction is missing a comma after "including the target population,". Otherwise, the sentence is too long and difficult to follow. - Page 4, line 17. Although it has been mentioned in the abstract and the "Strengths and limitations of this study" section, it is the first time that EHT appears in the introduction, and thus, it should be spelled out. - Page 4, line 40-44. This sentence is key to understand the relevance of EHTs, and the rationale for choosing them for this review. Still, it would be necessary to also include a definition of EHTs and some examples of what EHTs are, to help the reader understand what the authors mean by EHTs (smartphone apps to track and monitor chronic diseases? Wearable devices? Other type of technology?). - Page 5, line 25-27. In the aim of the study, it specifies the type of EHT "The types of EHTs considered for this review are those which provide patients with the ability to acquire information with the sole intent to support their healthcare decisions". However, this pre-requisite doesn't seem to be included in Table 2, eligibility criteria. Also, what other type of EHTs are there? - Considering that EHTs are the main focus of this review, more context is needed about them. In table 3, the authors mention that EHTs will be classified according to WHO's classification of digital health interventions (23). This reference seems highly relevant and should be mentioned in the introduction as well, as it seems the authors used this document as a reference not only to classify EHTs but also to define them. Methods

	- Page 5, line 41-42. This is the first time “collaborative design” is mentioned. Is co-design a short form to say “collaborative design”? If so, this should be introduced earlier, in the opening sentence of the introduction. - Page 6, line 45-46. Similarly to the level of participation and involvement processes, which were briefly explained, it would be very useful to establish which “types of healthcare tools” and “older adults’ needs” were available as options. These variables include references for the classification method used (23 and 24, respectively), but further explanation is needed. To better organize this information, a sub-section named “Outcomes” could be created within the methods, where all these variables (level of participation, involvement processes, frameworks, types of healthcare tools, older adults’ needs, …) are described in depth. - Page 7, lines 3-10. To better understand which levels of participation the authors had agreed upon, table 11 should be added here, without the second column. - Page 7, lines 10-14. The full name and hyperlink of Figure 1 is included here. The full name should be deleted from the text. - Page 7, lines 34-41. The title of this section does not correspond to the paragraph below it. The “study selection” section is associated to table 2, as it encompasses the eligibility criteria for the selection of the studies to be included. Therefore, it should have been presented earlier in the methods section. In that line, the paragraph below (lines 35-41) should not be part of the methodology, but the results instead, as the authors are reporting the main result of the systematic review (i.e., PRISMA flow diagram). Results - Page 7, line 49. The authors should reference table 4 early in the “Characteristics of Included Studies” section within the results. - Pate 7, line 50. “All studies indicated that older adults were involved in the development of an EHT.” Wasn’t this part of the inclusion criteria for the systematic review? As such, it was already expected that the included studies involved older adults, and the sentence is redundant. - Page 7, lines 51-53. The sentence “We were unable to obtain which co-design approach was used in 2 of the included studies (26,29), and 2 did not define a specific theoretical framework or co-design principle (8,30)” should be incorporated later in this section. The first paragraph should state the summary characteristics of the studies, included in table 4; and the co-design approaches and theoretical framework are not part of that. Furthermore, a results table summarizing the outcome variables for each study would be highly beneficial to better interpret and visualize the aforementioned sentence. - Page 8, line 14-17. There is no need to list all the countries where the studies took place. Readers can already see that in table 4. It is more interesting to report, by continent, where the studies are from. For instance, authors could include a sentence like the following: “N1% of the studies took place in North America, N2% in Europe, and only one study (N3%) was from Asia”. - Page 8, lines 18-24. Study quality was not reported in table 4 and therefore, this paragraph should be presented later in the results section, and it should have its own sub-section named “Risk of bias assessment” or “Study quality assessment”. - Page 8, line 36-37. “Of those that indicated a specific approach, 10 did not define the co-design approach which was utilized in the

	study (8,9,27,30–33,39,41,42).” Where does this information come from? This is not presented in Table 5. - Page 8, line 38-41. The numbers provided to describe the frequency of the co-design approaches do not match those from Table 5. Also, where are “involvement with end user/stakeholder, system usability, user needs, prototypes, and iterative develop” mentioned in table 5? Are these the definitions? What is the difference between a co-design approach and the definition? - Page 8, line 49. The first sentence should be second, and viceversa. It is better to present the topic of iterative development, and then mention the number of phases. Also, the number of phases is presented in table 6, and thus, it should be mentioned after table 6 is referenced in the text. - Page 9, line 11. The appropriate table should be referenced in this section. - Page 9, line 32. While there are many frameworks reported by the studies, since table 7 includes the frequencies, it might be noteworthy to mention what frameworks were more commonly reported (ISO Standard and IPDAS). - Page 9, lines 40-43. These sentences are repeated, as they were already stated in the characteristics of included studies. - Page 9, lines 45-47. This information could be included earlier in the results section, as part of the characteristics of included studies. While table 8 helps visualize the percentages of participant types, this could be simply mentioned in the text as it can be inferred from table 4. Thus, table 8 is unnecessary and could be deleted. - Page 9, lines 47-40. There is no need to list all the specific diagnoses, as they are already included in table 4. The sentence could be simplified to the following: “A total of 9 studies targeted participants with specific diagnoses (Table 4)”. - Page 10, line 7. Considering how the information is presented in table 9, it is hard to infer the sentence “Each study utilized between 2 to 11 involvement processes”, given that we can’t see how many processes each study reported. We would need a results table to visualize that. Instead, what we can observe in table 9 is the involvement processes that were most popular or mostly reported [prototype (medium fidelity) and interview (n=18), followed by prototype (functional) (n=15)] and the least popular/reported [remote evaluation (n=1)]. - Page 10, line 12. The corresponding table should be referenced here. Are there any more conclusions we can draft from the cross-tabulation of co-design approaches and iterative development? - Page 10, lines 40-49. Although the authors have properly referred to Figure 3, it is not clear which studies they are comparing when stating the percentage increases and decreases. Initially, from the first sentence, I would assume the authors are about to compare the studies with and without empowerment (first and second bars in the figure). However, it seems they are comparing the first bar with the third, fifth and seventh bars (Collaborate Yes, Involve Yes and Consult Yes). Why are they doing these comparisons? What is the purpose of doing that? Additionally, in the second sentence, they do seem to compare Empower Yes vs No, however they do not do that comparison for the other 4 levels of participation. Furthermore, they include table 12 as the cross-classification of co-designing approaches and levels of participation, but there are no comments about this table. What can we conclude from it?

	- Page 11, lines 7-29. This description of how the EHTs were classified should not be reported as a result, but as part of the methods instead, as per my second comment from the methods. - Page 11, lines 32-35. This information doesn't need to be listed in the paragraph, as it is already available in the table. - Page 12, lines 5-8. The categorization of older adults' needs should not be part of the results; it should be included within the methods - Page 12, lines 25-30. Given the criteria of including EHTs with the sole purpose of supporting healthcare decisions, is this sentence implying that the systematic review should actually include only those 14 articles, and not the 20 initially selected? This is very confusing. Discussion - Page 13, lines 24-27. This sentence is repeated, as it was also included in page 10, lines 32-36. - Page 13, line 27-29. "Of interest, is that within the studies that empowered participants, we observe an increase in the utilization …" This sentence needs grammar revision, and language could be simplified to: "Within the studies that empowered participants, …". Also, I am missing some comparisons to previous literature in this paragraph. - Page 13, line 40. After highlighting the need for a framework, do the authors have any recommendation based on what the results from this systematic review? Would they recommend any of the existing frameworks, or maybe the need for a new one that is more comprehensive? If so, please justify. - Page 15, lines 12-18. This sentence is hard to follow. I recommend changing the wording slightly to facilitate understanding. - Page 15, lines 23-33. These sentences are reporting results and thus, they should not be part of the discussion. Although some of this information is needed to contextualize the issue of measurement inconsistency, these sentences should be paraphrased and simplified. Also, more discussion around the inconsistency of user testing assessments should be incorporated; as it is written now, it is just a statement. Have other authors found the same issue? Is this finding common? Have possible solutions been considered? - Page 15, line 38. One of the biggest limitations of this study is the difficulty to extract and classify the information, although authors have used mitigation strategies (e.g., guides or already-stablished classification systems). This applies to most variables, from stakeholder involvement to EHTs. Tables - Table 3 – Study characteristics. The two options "sample size" and "number of older adults" seem redundant. I assume that sample size corresponds to the total number of participants in the study, while number of older adults refers specifically to the target population of this review. I understand the authors are trying to reflect that these numbers are not necessarily the same, but just to clarify, it would be better to express this the following way: "study sample size" or "sample size of the study". - Tables 5 and 6 could be combined. This will help reduce the number of tables/figures the manuscript has. - Tables 9 and 10 could be also combined, by adding one more column on the right in table 10 that includes the "total studies n=25" and the percentages. The references could be added as

	hyperindex, next to the corresponding number, similarly to tables 12 and 15. - Tables 11 and 12 could be combined, by adding one more line at the bottom of table 12 with the total number of studies. - Tables 14 and 15 could be combined, by adding one more line at the bottom of table 15, including the total number of studies - Table 16 has an extra line that should be deleted, right before non-validated objective measures. It confuses the reader when trying to understand the classification presented in the table. Also, "Cognitive process" should be not bolded. - In general, the number of tables is quite high. As per BMJ Open's submission guidelines, the recommended number of figure and tables is "up to five", although "this is flexible, but exceeding this will impact upon the paper's 'readability'." Even if the journal allows more than 5 tables and figures, this manuscript presents 16 tables and 3 figures and most of them could be deleted or combined. While some of the tables are necessary (e.g., tables 2 and 4), others could be included as supplementary material (e.g., tables 1 or 3). Additionally, tables 5-9, 11, 13, 14 and 16 are describing each one of the review results individually (co-design approaches, iterative development, theoretical framework, study participants, co-design involvement processes, levels of participation, types of EHTs, older adults' needs, and subjective and objective measures of user testing) and some of these could be combined. Typically, in a systematic review, there are two key tables that are similar in structure: the study characteristics table and the results/outcomes table. Both of these have all the studies in the first column, one per row, and the other columns include the different variables observed for study characteristics, and results/outcomes, respectively. Although the individual tables provide important information regarding the frequency of articles that report each variable and this is helpful towards writing the results section, it makes most sense to present the majority of this information in one table (when possible), and keep the individual tables as supplementary material too. For example, the authors could add a results table including the main categories presented in table 3 (co-design approaches, stakeholder involvement and electronic healthcare tools). Lastly, in regards to tables 12 and 15 that cross-tabulate two variables, authors should consider whether the information presented is relevant and worth-mention. What can we infer from these tables? Is the information useful? Does it help to make any associations relevant for the review? Although these tables could elucidate and portray key information and associations between variables, if the answer the previous questions is no, then it is not necessary to include the tables in the manuscript. References - Reference 6 and 43 are not properly referenced; they all should include the website name and the date it was accessed. - Reference 10 is missing information. If it's an article, it is missing the journal name, volume, pages and year of publication - In reference 19, a website link could be added - References 23 and 25 should include the date it was accessed, not cited.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Marleen de Mul, Erasmus Universiteit Rotterdam Erasmus School of Health Policy and Management Comments to the Author:

Overall comment: Understanding how co-design works in practice for different sub-groups is a relevant topic. The researchers have followed a well-documented, thorough process to select papers for their review, with the aim of providing further insight into co-design. I appreciate that the authors also assessed the quality of the included papers. However, I am puzzled by the purpose of this particular review, and how the result of this study adds additional insights to Fischer's much-cited review. The aim is promising, but for me the analysis does not yet meet my expectations. This may have to do with the way the results are presented, and the choice of counting over narrative description. Counting how many different frameworks there are, or what the average number of phases is in co-design projects with older adults, is -in itself - not very interesting to me. It does not help me to better understand the specifics, the challenges, and the work required to create value from co-design with older people. I advise the authors to analyse their selected articles in other ways and also to present the data they have found in a different way. This will make the study more relevant to the readership of BMJ Open, who are not designers and therefore lack the ability to pick up on the nuances of co-design terminology.

Response to the overall comment: We are grateful for the encouraging comments and inspiring us to look further into Fischer's review. We have now reassessed our review, created a table (see below) to compare the similarities between our and Fisher's review and substantiate the differences to highlight where our review provides additional value to the readership of BMJ Open. By drawing these comparisons, it also gave us the opportunity to further reflect on additional insights we had gathered from our included studies, specifically related to learning outcomes. We believe it was necessary to mechanically extract data from each study to determine whether there were patterns in the specific involvement/participation processes and testing measures associated with specific learning outcomes. As this review focuses on the development of EHTs, specifically the tools used to inform healthcare decisions, evaluating the role of learning through co-design approaches is necessary. If learning does not occur to the extent necessary to make qualified decisions, the information generated for consumption by the end-user may not be validated. We hope that the reviewer finds this new focus of our manuscript value-added.

	Fisher et al.	**Cole et al.**
Published between	2014-2018	2010-March 2021
Main Consequences	Learning, Adjusted Design, Increased sense of participation	Usability, Errors, Performance, Need for Assistance, Meaningful experience-mutual learning (e.g., who is learning?)

User Involvement	Structured process whose outcomes are contingent on a range of premises.	User involvement is focused on whether the tool is improved.
	Synthesized results. Develop concept of user involvement. Present analytical framework.	Synthesized results. Identify gaps in mutual learning. Propose need for improving co-design frameworks.
Main Questions	1. What purposes are described in the literature for involving older people in the design of a technology? 2. How are older people selected for involvement in design practice? 3. What roles are assigned to older people when they are involved? 4. **At what design stages are older people usually involved?** 5. **At what levels of involvement are older people usually involved?** 6. What images of older people prevail in the literature involving older people? 7. What consequences of involving older users are re-ported in the literature?	1. What approaches are used for involving older adults in developing EHTs? 2. How are these approaches defined? 3. What frameworks were utilized? 4. Was iterative development utilized? How many phases? 5. **What stages of development are the approaches employed?** 6. **What is the level of involvement for older adults?** 7. What involvement processes are employed? 8. What are the subjective and objective measures used? 9. Are the subjective and objective measures validated? -------- REVISIONS NOW INCLUDE 10. Did participants reflect on testing measures? (ie., why did they make errors?) 11. Was there a bi-directional flow of knowledge between participants and researchers?
Levels of involvement	Low, Intermediate, High Arnstein (1969)	Inform, Consult, Involve, Collaborate, Empower Vaughn (2020)
Search Strategy	broad terms (a) “user involvement” and (b) “older adults.”	"co-design", "aged", "**electronic healthcare tools**",
Databases	Scopus, Web of Science	PubMed, Embase, Scopus

Analyzed	• Thematic Analysis, Coding • Differently reported consequences of • User involvement • Purposes • Stages and levels of involvement • Roles and images of older peoples • Selection procedures	• Synthesized results • How are approaches used? • How approaches are classified? • Terminologies • Frameworks • Stages of Development • Involvement processes • Levels of Participation • User testing measures ---- REVISIONS NOW INCLUDE • Learning Levels • Learning Beneficiary (Who is Learning)
User Involvement of Older People—How Does It Matter?	1. Learning, 2. Adjusted Design, and 3. Improved sense of participation. We also found consequences of older user involvement for which the literature is ambiguous	1. Learning - Bidirectional knowledge 2. Improvement in participation 3. Improved engagement in healthcare decisions
	First, from the empirical studies we reviewed, the effect of involving older users on technology ***acceptance*** remains unclear. **Second**, we found that the literature is not clear about the impact of older user involvement on technology ***adoption***.	Studies that empowered participants were more likely to utilize specific involvement processes such as contextual inquiry and functional prototyping. **Learning outcomes: *Mutual learning*** occurs when participants, designers and researchers learn from each other, or when participants learn from **reflecting** on their own efforts and errors during user testing. Learning can also occur when participants are **taught or exposed to technology prior to user testing** .
	User involvement can have different outcomes, and the type of outcomes achieved appears to differ with respect to the way designers choose to involve older people.	Learning outcomes may vary based on the types of involvement processes.
	Encourage further research on the relationship with acceptance and adoption, as well as on the actual practices of user involvement.	To ensure studies are optimizing the involvement of older adults when developing EHTs, improving existing codesign frameworks to meet the needs of older adults should be evaluated.

Suggestion 1: At present, the emphasis in the first part of the results is on distinguishing the methodological terms used by the authors of the original studies. Without clarification of what these terms mean and, for example, ordering these terms into classes, it remains a random (alphabetical) list and there is, for example, no clear substantive difference between table 5 and tables 6 and 7. It is unclear what a reader should do with this information. Table 9 is informative, but cross-table 10 not so much. What do they show? I recommend the current Table 4 be expanded, to include information per study on the co-design approach chosen, before the aggregate data is presented in later tables. If the co-design methodology is and will remain a central theme in the paper, I am interested not only in the aggregated data, but also in the original data.

Response to suggestion 1: Thank you for this recommendation. We have extended the summary characteristics table (Table 4 in the original submission). This table now includes columns for Co-design Approach, Co-design Approach Defined, and No. of Phases. This allowed us to eliminate Table 5 (CoDesign Approaches and Definitions) and Table 6 (Iterative Development compared against Co-Design terminology). We have also redesigned the objective of this review and focused on learning outcomes, therefore we added two columns, Learning Level and Learning Beneficiary. By incorporating this information into the characteristics table, readers are now able to view the original data for each study, rather than the aggregated results presented in the original tables. We reshaped our review and focused our attention on learning outcomes from each study. Learning outcomes are threaded throughout the manuscript to illustrate their relationship with the original extracted variables and are now the focal point of the review. Based on your recommendations and those of Reviewer #2, we have made significant revisions to our tables (see below).

Original Table Number	**New Table Number**	**Table Revisions**
Table 1 - Search Results	Supplementary Appendix A – Search Results	This table has been changed to a supplementary table.
Table 2 - Inclusion and Exclusion Criteria	Table 1 - Inclusion and Exclusion Criteria	No changes
Table 3 - Study Characteristics	Supplementary Appendix B- Study Characteristics	Adjusted the name of one variable, from "Sample size" to "Number of Stakeholders". This table has been changed to a supplementary table.
Table 4- Summary Characteristics	Table 3 - Summary Characteristics	This table now includes columns for Co-design Approach, Co-design Approach Defined, and No. of Phases. This allowed us to eliminate Table 5 (Co-Design Approaches and Definitions) and Table 6 (Iterative Development compared against CoDesign terminology). We have also redesigned the objective of this review and focused on learning outcomes, therefore we added two columns, Learning Level and

		Learning Beneficiary. For clarity, we adjusted "sample size" to "No. of Stakeholders".
Table 5 – Co-Design Approaches and Definition	Removed Table	This table has been removed from the manuscript. The co-design approach of each study is now included in Table 3.
Table 6 – Iterative Development compared against Co-Design terminology	Removed Table	This table has been removed from the manuscript. The number of phases of each study is now included in Table 3. We did not draw insights from comparing iterative development against codesign terminology, and as our review now focuses on learning outcomes, we have removed this table.
Table 7 – Theoretical framework and Co-Design Principles	Removed Table	This table has been removed from the manuscript. We have included a supplementary table that displays which frameworks were utilized by each study.
Table 8 – Study Participants	Removed Table	This table has been removed from the manuscript. This information already existed in the Summary Characteristics table, which is now Table 3.
Table 9 – Co-Design Involvement Processes	Table 6 - Co-Design Involvement Processes by Learning Levels	This table was initially used to summarize the number of studies that utilized specific involvement processes. We are now using this table to cross-classify the learning levels with the specific involvement processes. This was important for us to distinguish the specific

		processes that were used more frequently in the studies which were classified with higher levels of learning.
Table 10 – Co-Design Approaches cross-classified with Co-Design Involvement Processes (color-coded to reflect association in Figure 3)	Removed Table	This table has been removed from the manuscript. We are no longer trying to extract insight from this cross-classification.
Table 11 - Levels of Participation	Table 2 - Levels of Participation	We introduce this table earlier in the manuscript to provide Vaughn's definitions of the 5 levels of involvement. We removed the "Total studies" column, as this did not provide value.
Table 12 - Cross-classifying codesign approaches with levels of participation	Supplementary Appendix D - Cross-classifying co-design approaches with levels of participation	This table has been changed to a supplementary table.
Table 13 –WHO Classification of Digital Health Interventions	Table 4 –WHO Classification of Digital Health Interventions	No changes
Table 14-Older adults' needs addressed by electronic healthcare tools	Removed Table	This table has been removed from the manuscript. We have redesigned the review and are no longer categorizing older adults' needs.
Table 15 - Cross-classification of older adults' healthcare technology needs with the WHO classification of digital health interventions	Removed Table	This table has been removed from the manuscript. We have redesigned the review, and are no longer categorizing older adults' needs, therefore there is no need to cross-classify with the classification of EHTs.

Table 16 – User Testing – Subjective and Objective Measures	Table 5 – User Testing – Subjective and Objective Measures	No changes were made to the information presented; however, adjustments were made to formatting and typographical errors.
	Supplementary Appendix C	We established a results table for theoretical frameworks, specific involvement processes, user testing (endpoints, measures and validation)

Suggestion 2: You could analyse the selected papers for working mechanisms, how the authors wrote about the involvement of older people in different phases of their project. Was it different from other target groups? How did the researchers specifically take into account the characteristics of older people such as lower health literacy, anxiety for technology, endurance… when deciding where to involve older adults and where not, or when deciding to use tool A rather than tool B. This requires a different analysis of the text, going beyond what can be counted and averaged.

Response to suggestion 2: We appreciate the guidance and suggestion to further analyze our included studies from the perspective of how older people are involved in the different phases of the project, and whether their involvement was different from other target groups. We believe evaluating this perspective as well as additional characteristics such as lower health literacy, anxiety for technology and endurance are great suggestions. We found it difficult to extract the specific involvement processes and the associated characteristics expressed during specific phases of the project, therefore were unable to include this in our revised review. However, we feel this would be valuable to explore in future research efforts. We chose to take a modified (per reviewers' valuable suggestions) approach to our review and focused on extracting learning outcomes. We do feel there is relevance in evaluating whether mutual learning occurs when engaging older adults in the development process, and if it does, whether it occurs more frequently when engaged in specific involvement processes. We found that current frameworks do not reference mutual learning, nor do they suggest that a learning model should be employed when co-designing with older adults. However, evidence extracted from the studies suggests that the involvement processes utilized may be related to the levels at which stakeholders participate and can either inhibit or encourage mutual learning. We also found that for studies in which participants and researchers benefitted from mutual learning, we observed a higher utilization of specific involvement processes including contextual inquiry, cognitive walkthroughs, focus groups, working group workshops and functional prototyping, and a decrease in think aloud sessions

Suggestion 3: Critically review the classifications:
a. The study by Herrmann et al. is used to classify the needs of older adults addressed by EHTs. I wonder if it is relevant to use an additional classification for the EHT included in the evaluation (on top of the WHO classification), as this is not the main subject of the evaluation. Another reason why combining these two frameworks is less valuable is that both frameworks relate to care innovation in general, not specifically to older people. In my opinion it would be sufficient to use either WHO or Herrmann and not both. We know that there are many more studies on EHT, some even specifically for older people. However, they were not included in this review because no co-design methodology

was described. The type of EHT is not a relevant result, but a characteristic of the selected papers. Therefore, I would add this information to the first section of the results or an appendix, not to the last section (implying that it is the climax of the study). The cross-classification in Table 15 is not relevant for the purpose of this article.

Response to suggestion 3a: Thank you for bringing this to our attention. We agree and have removed the classification of older adults' needs, which also means we have removed the cross-classification.

Suggestion 3b. I like the idea of going deeper into how, when and how intensively the older adults are involved. However, I think it would be of value to first apply Vaughn's model to the different co-design methodologies (for example, in human-centered design the role of the user in development is different than in co-creation). In Sanders & Stappers' book, there is also a matrix that positions different design methodologies in terms of user involvement. This would make the decision on which level to place a particular paper more transparent and replicable. Table 12 and the text on page 10, lines 41-49 are not clear.

Response to suggestion 3b:
Through our revised approach to this review, we chose to move Table 12- Cross-classifying co-design approaches with levels of participation to the appendix as a supplementary table. This table was created to analyze whether there were relationships between the co-design approach and the levels of participation. Findings were limited to the levels of participation, and involvement processes, therefore further cross-classification of these variables was not conducted, as we were unable to specify the point within the study that a particular involvement process was employed. We appreciate your suggestion to look further into both Vaughn's model and how this relates to the different co-design methodologies. We found similar difficulties in drawing insights around these relationships but feel this is a great suggestion and one that we would like to explore in future research. We also investigated incorporating Sanders & Stappers matrix, which would allow us to position each approach in terms of user involvement. We again found that our studies were spread across a plethora of approaches and that positioning them within this matrix did not provide us with the ability to draw further insights.

Other comments:
Looking at the search string, I wondered if papers on personal health records and apps were also retrieved. These also seem like important tools to inform healthcare decisions.

Response: Yes, our included studies consisted of both mobile and web applications and three studies (Kim et al, Ahmed et al, Mansson et al) developed tools that gave access to their personal health records.

Minor comments:
#1: The section on strengths and limitations seems difficult to read, due to incomplete sentences and lack of signal words
Response to minor comment #1: Thank you for suggesting we make revise this section. We revised this text to reflect complete sentences.

- This review provides a comprehensive assessment of the utilization of co-design approaches in studies that involve older adults in developing electronic healthcare tools.
- We developed a broad search strategy to ensure all co-design or "collaborative" approaches were captured in this review.
- We utilized widely reported classification methods to categorize the included studies.

- We limited inclusion criteria to studies that involved older adults in the development of an electronic healthcare tool (EHT).
- We limited our analysis to the overall involvement processes and levels of participation at the study level; cross-classification of these variables at the design stage level for each study was not conducted.

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#2: Add a column with the reference number in Table 4.
Response to minor comment #2: All references have been added to the summary characteristics table.

#3: The headings could also be more self-explanatory as this is not a review of a clinical study. "No stakeholders" makes more sense than "sample size".
Response to minor comment #3: Thank you for this recommendation. We adjusted Sample Size to No. of Stakeholders.

#4: References 47 and 51 are the same.
Response to minor comment #4: Thank you for finding this error on our part. We have adjusted the reference.

Reviewer: 2
Dr. Lucía Otero Varela, University of Calgary Comments to the Author:

Overall comment: This is a systematic review on "Co-design approaches involving older adults in the development of electronic healthcare tools", submitted to BMJ Open by Cole et al. The authors have thoroughly followed a rigorous methodology to synthesize the current state of the literature on a novel area that involves a growing population (adults aged 60 and older). Adapting electronic healthcare tools to meet their needs is imperative, and thus, the topic of this review is relevant and interesting to the BMJ Open audience, which makes this manuscript a worthwhile piece of translational work.

Nonetheless, although the complexity of this topic should be acknowledged, there is potential for improvement, especially in terms of reporting. The authors summarize the manuscript appropriately in the abstract and provide a comprehensive introduction to contextualize the focus of the review. However, the methods must be better described, placing focus in the outcomes and data synthesis, and the results need to be clearly presented. Special attention should be given to the tables (too numerous): firstly, there are too many, some should be combined or deleted; secondly, the results table is missing, which is key to understand the findings from this review; and thirdly, there are tables from which no inferences are drawn, thus, the authors should reflect on whether these were necessary or should be deleted. For detailed recommendations, please see my comments below, categorized by manuscript section.

Response to the overall comment: Thank you for the kind words; and recognition of our hard work on this manuscript. We agree with your overall impression of our original submission. We have deleted and combined tables to reduce our table count, expanded the characteristics table and created supplementary files for additional study results. Based on comments from you and reviewer #1, we have created a table to detail the table changes, including the reason why we made each adjustment.

Introduction

Suggestion #1: Page 3, line 33. The opening sentence of the introduction is missing a comma after "including the target population,". Otherwise, the sentence is too long and difficult to follow. **Response**

to suggestion #1: We have revised the sentence “Co-design approaches aim to elicit ideas and foster a non-hierarchical environment for stakeholders, including the target population, in which their concepts, tacit knowledge, and lived experiences can be applied to develop tools that meet their needs, improve usability and impact outcomes from use of the system(1,2).”

Suggestion #2: Page 4, line 17. Although it has been mentioned in the abstract and the “Strengths and limitations of this study” section, it is the first time that EHT appears in the introduction, and thus, it should be spelled out.
Response to suggestion #2: Thank you for catching this, we have now spelled out electronic healthcare tools.

Suggestion #3: Page 4, line 40-44. This sentence is key to understand the relevance of EHTs, and the rationale for choosing them for this review. Still, it would be necessary to also include a definition of EHTs and some examples of what EHTs are, to help the reader understand what the authors mean by EHTs (smartphone apps to track and monitor chronic diseases? Wearable devices? Other type of technology?).
Response to suggestion #3: We have now provided definition and examples. “We are using the term
EHT to encapsulate all terms classified and defined by the World Health Organization (WHO) Digital Health Interventions (26). EHTs are interactive web applications, mobile applications or wearables that track or provide guidance on relevant healthcare issues to improve wellbeing. For instance, monitoring health risks (i.e., risk of falling, risk of heart failure), or personal health information management (i.e., patient decision aids, access to health records and educational materials).”

Suggestion #4: Page 5, line 25-27. In the aim of the study, it specifies the type of EHT “The types of EHTs considered for this review are those which provide patients with the ability to acquire information with the sole intent to support their healthcare decisions”. However, this pre-requisite doesn’t seem to be included in Table 2, eligibility criteria. Also, what other type of EHTs are there?
Response to suggestion #4: Thank you for bringing this to our attention. We have deleted this sentence from this section of our review, as it does not provide clarity or value.

Suggestion #5: Considering that EHTs are the main focus of this review, more context is needed about them. In table 3, the authors mention that EHTs will be classified according to WHO’s classification of digital health interventions (23). This reference seems highly relevant and should be mentioned in the introduction as well, as it seems the authors used this document as a reference not only to classify EHTs but also to define them.
Response to suggestion #5: Thank you for noting the relevance of this classification. Along with defining EHTs, we are now referencing the WHO in the introduction, per your suggestion.

Methods

Suggestion #6: Page 5, line 41-42. This is the first time “collaborative design” is mentioned. Is co-design a short form to say “collaborative design”? If so, this should be introduced earlier, in the opening sentence of the introduction.
Response to suggestion #6: We removed the word “collaborative” and reframed the sentence to “A broad search strategy was utilized to ensure all approaches that engage end users in the design of EHTs were captured in this review.”
Suggestion #7: Page 6, line 45-46. Similarly to the level of participation and involvement processes, which were briefly explained, it would be very useful to establish which “types of healthcare tools” and “older adults’ needs” were available as options. These variables include references for the classification method used (23 and 24, respectively), but further explanation is needed. To better organize this information, a sub-section named “Outcomes” could be created within the methods,

where all these variables (level of participation, involvement processes, frameworks, types of healthcare tools, older adults' needs, …) are described in depth.
Response to suggestion #7: Thank you for this suggestion, we agree this would have allowed for better organization of our previously extracted information. Our revised manuscript now focuses on Learning Outcomes, which we have described in-depth and now thread throughout the manuscript.

Suggestion #8: Page 7, lines 3-10. To better understand which levels of participation the authors had agreed upon, table 11 should be added here, without the second column.
Response to suggestion #8: We moved the table to the corresponding section and removed the second column that indicated the total number of studies. We now reference the table in this section of the manuscript.

Suggestion #9: Page 7, lines 10-14. The full name and hyperlink of Figure 1 is included here. The full name should be deleted from the text.
Response to suggestion #9: Thank you for catching this, it was an oversight and we have now removed the full name of hyperlink.

Suggestion #10: Page 7, lines 34-41. The title of this section does not correspond to the paragraph below it. The "study selection" section is associated to table 2, as it encompasses the eligibility criteria for the selection of the studies to be included. Therefore, it should have been presented earlier in the methods section. In that line, the paragraph below (lines 35-41) should not be part of the methodology, but the results instead, as the authors are reporting the main result of the systematic review (i.e., PRISMA flow diagram).
Response to suggestion #10: We removed the study selection heading and moved the systematic review text to results section.

Results

Suggestion #11: Page 7, line 49. The authors should reference table 4 early in the "Characteristics of Included Studies" section within the results.
Response to suggestion #11: We now reference the Summary Characteristics table at the beginning of this section.

Suggestion #12: Page 7, line 50. "All studies indicated that older adults were involved in the development of an EHT." Wasn't this part of the inclusion criteria for the systematic review? As such, it was already expected that the included studies involved older adults, and the sentence is redundant. **Response to suggestion #12:** We have removed this statement.

Suggestion #13: Page 7, lines 51-53. The sentence "We were unable to obtain which co-design approach was used in 2 of the included studies (26,29), and 2 did not define a specific theoretical framework or co-design principle (8,30)" should be incorporated later in this section. The first paragraph should state the summary characteristics of the studies, included in table 4; and the co-design approaches and theoretical framework are not part of that.
Response to suggestion #13: We have moved this state to the co-design approaches section.

Suggestion #14: Furthermore, a results table summarizing the outcome variables for each study would be highly beneficial to better interpret and visualize the aforementioned sentence.
Response to suggestion #14: We added two columns (1) co-design approach and (2) co-design approach defined to the summary characteristics table.

Suggestion #15: Page 8, line 14-17. There is no need to list all the countries where the studies took place. Readers can already see that in table 4. It is more interesting to report, by continent, where the

studies are from. For instance, authors could include a sentence like the following: “N1% of the studies took place in North America, N2% in Europe, and only one study (N3%) was from Asia”.
Response to suggestion #15: We have revised the text based on recommendation.

Suggestion #16: Page 8, lines 18-24. Study quality was not reported in table 4 and therefore, this paragraph should be presented later in the results section, and it should have its own sub-section named “Risk of bias assessment” or “Study quality assessment”.
Response to suggestion #16: We have revised the heading to “Study Quality Assessment” and moved this to the beginning of the results section.

Suggestion #17: Page 8, line 36-37. “Of those that indicated a specific approach, 10 did not define the co-design approach which was utilized in the study (8,9,27,30–33,39,41,42).” Where does this information come from? This is not presented in Table 5.
Response to suggestion #17: We added a column in the Summary Characteristics table to indicate whether each study defined their approach.

Suggestion #18: Page 8, line 38-41. The numbers provided to describe the frequency of the co-design approaches do not match those from Table 5.
Response to suggestion #18: We revised this information and moved to Table 3-Summary Characteristics.

Suggestion #19: Also, where are “involvement with end user/stakeholder, system usability, user needs, prototypes, and iterative develop” mentioned in table 5? Are these the definitions? What is the difference between a co-design approach and the definition?
Response to suggestion #19: We have revised the objective of this review and are placing less focus on definitions. Table 5-Co-Design Approaches and Definition has been removed from the manuscript. We are now listing the co-design approach of each study in Table 3-Summary Characteristics. We included a summary of definitions, “Reviewing the definitions of co-design approaches, 8 studies refer to involvement with end-user/stakeholder, 5 studies refer to system usability, 34 studies refer to user needs, 3 studies refer to using prototypes, and 3 studies refer to iterative development. Studies did not determine which involvement processes were used when defining the approach.” We also provided this key finding, “While 13 definitions were represented across the studies, general themes presented suggest that the approaches could be used interchangeably. While this may further exacerbate a problem, and lead to inconsistent use of terminology, what is of notable importance within this review, is not so much on the name of the approaches used, but how they are interpreted and then employed to determine the methods for involving and co-designing with older adults.”

Suggestion #20: Page 8, line 49. The first sentence should be second, and viceversa. It is better to present the topic of iterative development, and then mention the number of phases. Also, the number of phases is presented in table 6, and thus, it should be mentioned after table 6 is referenced in the text. **Response to suggestion #20:** Thank you for this recommendation, we have revised this text to now read. “Table 5 includes co-design approaches, number of phases conducted in each study and whether they defined their approach. Overall, 9 different co-design approaches were represented among the 25 studies. Of the 25 studies, 7 studies indicated the use of more than one co-design approach
(8,9,38,41,43,45,46), and 2 studies stated that stakeholders were involved in development but did not indicate a specific approach (29,39). Overall, 19 of the 25 studies conducted their study in multiple phases, ranging between 2 to 6 phases, and an overall mean of 3.4 phases per study. Of the 6 studies which did not employ iteration in development, 2 labeled user involvement as either a human factors or usability evaluation approach and did not provide definitions to the approach utilized (34,45). Reviewing the definitions of co-design approaches, 8 studies refer to involvement with end-

user/stakeholder, 5 studies refer to system usability, 34 studies refer to user needs, 3 studies refer to using prototypes, and 3 studies refer to iterative development. Studies did not determine which involvement processes were used when defining the approach."

Suggestion #21: Page 9, line 11. The appropriate table should be referenced in this section.
Response to suggestion #21: We have expanded this information and created a supplementary table
(Appendix A) that now includes the full list of theoretical frameworks or co-design principles employed for each study. We are now referencing Appendix A in this section. "Appendix A summarizes the key theoretical frameworks or co-design principles employed for each study."

Suggestion #22: Page 9, line 32. While there are many frameworks reported by the studies, since table 7 includes the frequencies, it might be noteworthy to mention what frameworks were more commonly reported (ISO Standard and IPDAS).
Response to suggestion #22: Thank you for this suggestion, we have now added a statement as to which frameworks were most commonly reported. "The most commonly reported frameworks were Human Centered Design Methodology (ISO Standards) (n=5), International Patient Decision Aid Standards (IPDAS) (n=4), and User Center Design (n=3)."

Suggestion #23: Page 9, lines 40-43. These sentences are repeated, as they were already stated in the characteristics of included studies.
Response to suggestion #23: Thank you for this suggestion, we have removed the redundant text under participant types.

Suggestion #24: Page 9, lines 45-47. This information could be included earlier in the results section, as part of the characteristics of included studies. While table 8 helps visualize the percentages of participant types, this could be simply mentioned in the text as it can be inferred from table 4. Thus, table 8 is unnecessary and could be deleted.
Response to suggestion #24: As suggested, we moved the suggested text to the characteristics of included studies section, and removed Table 8- Study Participants.

Suggestion #25: Page 9, lines 47-40. There is no need to list all the specific diagnoses, as they are already included in table 4. The sentence could be simplified to the following: "A total of 9 studies targeted participants with specific diagnoses (Table 4)".
Response to suggestion #25: We have adjusted the text based on your recommendation.

Suggestion #26: Page 10, line 7. Considering how the information is presented in table 9, it is hard to infer the sentence "Each study utilized between 2 to 11 involvement processes", given that we can't see how many processes each study reported. We would need a results table to visualize that.
Response to suggestion #26: We have added a supplementary table (Appendix A) that now includes the specific involvement processes employed by each study.

Suggestion #27: Instead, what we can observe in table 9 is the involvement processes that were most popular or mostly reported [prototype (medium fidelity) and interview (n=18), followed by prototype (functional) (n=15)] and the least popular/reported [remote evaluation (n=1)].
Response to suggestion #27: Thank you for this suggestion, we have incorporated the suggested text.
"A variety of involvement processes were used in the 25 studies, as indicated in Table 6, which reports 152 processes across all the included studies. Appendix A includes which involvement processes were used within each study. The involvement processes that were most widely reported were interviews and prototypes (medium fidelity) (n=18), followed by prototype (functional) (n=15) and

the least reported was remote evaluation (n=1). Each study utilized between 2 to 11 involvement processes, with a mean of 6 processes per study."

Suggestion #28: Page 10, line 12. The corresponding table should be referenced here. Are there any more conclusions we can draft from the cross-tabulation of co-design approaches and iterative development?
Response to suggestion #28: This table has been removed from the manuscript. The number of phases of each study is now included in Table 3-Summary Characteristics. We did not draw insights from comparing iterative development against co-design terminology, and as our review now focuses on learning outcomes, we have removed this table.

Suggestion #29: Page 10, lines 40-49. Although the authors have properly referred to Figure 3, it is not clear which studies they are comparing when stating the percentage increases and decreases. Initially, from the first sentence, I would assume the authors are about to compare the studies with and without empowerment (first and second bars in the figure). However, it seems they are comparing the first bar with the third, fifth and seventh bars (Collaborate Yes, Involve Yes and Consult Yes). Why are they doing these comparisons? What is the purpose of doing that? Additionally, in the second sentence, they do seem to compare Empower Yes vs No, however they do not do that comparison for the other 4 levels of participation.
Response to suggestion #29: We have revised this analysis and focused on comparing studies that reached the level of empower versus those which did not. "Figure 3 depicts each level of participation and the percent breakdown of associated involvement processes. The most distinct comparison is between the studies classified as Level 5-Empower (No) versus Empower (Yes). This comparison reflects a decrease in participatory design (from 69% to 55%) and an increase in functional prototypes (from 2% to 14%), respectively."

Suggestion #30: Furthermore, they include table 12 as the cross-classification of co-designing approaches and levels of participation, but there are no comments about this table. What can we conclude from it?
Response to suggestion #30: Table 12 has been changed to a supplementary table, as we are not drawing substantial conclusions from the table but wanted to offer this information to our readers.

Suggestion #31: Page 11, lines 7-29. This description of how the EHTs were classified should not be reported as a result, but as part of the methods instead, as per my second comment from the methods. **Response to suggestion #31:** We have moved the description of EHTs to the methods section.

Suggestion #32: Page 11, lines 32-35. This information doesn't need to be listed in the paragraph, as it is already available in the table.
Response to suggestion #32: We have removed the summary text based on recommendations.

Suggestion #33: Page 12, lines 5-8. The categorization of older adults' needs should not be part of the results; it should be included within the methods
Response to suggestion #33: We have revised the objective of this review, and are no longer assessing older adults' needs, therefore have removed this section from the review.

Suggestion #34: Page 12, lines 25-30. Given the criteria of including EHTs with the sole purpose of supporting healthcare decisions, is this sentence implying that the systematic review should actually include only those 14 articles, and not the 20 initially selected? This is very confusing.
Response to suggestion #34: We have revised the objective of this review, and concur that this sentence does not provide value, therefore we have removed the sentence.

Discussion

Suggestion #35: Page 13, lines 24-27. This sentence is repeated, as it was also included in page 10, lines 32-36.
Response to suggestion #35: Thank you for bringing this to our attention, this sentence has been removed.

Suggestion #36: Page 13, line 27-29. “Of interest, is that within the studies that empowered participants, we observe an increase in the utilization …” This sentence needs grammar revision, and language could be simplified to: “Within the studies that empowered participants, …”. Also, I am missing some comparisons to previous literature in this paragraph.
Response to suggestion #36: This statement did not provide value to our review, and we have removed it from the manuscript.

Suggestion #37: Page 13, line 40. After highlighting the need for a framework, do the authors have any recommendation based on what the results from this systematic review? Would they recommend any of the existing frameworks, or maybe the need for a new one that is more comprehensive? If so, please justify.
Response to suggestion #37: We have revised this text, “This need is further exemplified as 25% (n=6) of the included studies suggested recommendations for a framework that could be used when engaging older adults in the development of EHTs. Further research is needed to determine whether the proposed, or a combination of existing and proposed frameworks can be validated to optimize older adults’ involvement, one that provides an environment for mutual learning and empowerment.”

Suggestion #38: Page 15, lines 12-18. This sentence is hard to follow. I recommend changing the wording slightly to facilitate understanding.
Response to suggestion #38: We have revised the text under the Electronic Healthcare tools (EHTs) heading.

Suggestion #39a: Page 15, lines 23-33. These sentences are reporting results and thus, they should not be part of the discussion.
Response to suggestion #39a: We have moved this text to the results section.

Suggestion #39b: Although some of this information is needed to contextualize the issue of measurement inconsistency, these sentences should be paraphrased and simplified. Also, more discussion around the inconsistency of user testing assessments should be incorporated; as it is written now, it is just a statement. Have other authors found the same issue? Is this finding common? Have possible solutions been considered?
Response to suggestion #40a: Thank you for suggesting we investigate this further. We have summarized this text and found further evidence to support this inconsistency. “Furthering the notion that a framework, specific to older adults in developing EHTs, needs to be evaluated, is concerning the inconsistency of user testing assessments. We extracted evidence that 19 different measures were utilized in 76% (n=19) of the included studies and found that only five studies (20% of all included studies) used a validated subjective measure to evaluate their EHT. Usability was the most specified measure, and while the SUS, a validated usability assessment was utilized, more than half of the studies utilized a non-validated Likert-scale assessment. This discrepancy in assessments for user testing makes it difficult to draw definitive conclusions as to whether studies are consistently measuring what they intended to measure. This is consistent with a recent review that found studies often use non-validated questionnaires rather than readily available validated assessments and that despite recommendations to measure learning, this is rarely employed(47).”

Suggestion #40: Page 15, line 38. One of the biggest limitations of this study is the difficulty to extract and classify the information, although authors have used mitigation strategies (e.g., guides or alreadystablished classification systems). This applies to most variables, from stakeholder involvement to EHTs. **Response to suggestion #40:** We agree. We have now acknowledged this limitation to our revised manuscript. We also hope that our revised manuscript addressed to some degree this limitation.

Tables

Thank you for evaluating this review and making these great suggestions for improvements. We have created a table that highlights which tables we have consolidated and/or removed and included the specific changes we have made with each table.

Suggestion #41: Table 3 – Study characteristics. The two options "sample size" and "number of older adults" seem redundant. I assume that sample size corresponds to the total number of participants in the study, while number of older adults refers specifically to the target population of this review. I understand the authors are trying to reflect that these numbers are not necessarily the same, but just to clarify, it would be better to express this the following way: "study sample size" or "sample size of the study".
Response to suggestion #41: Thank you for this recommendation. We have taken your suggestion and the suggestion of reviewer #1, and adjusted "sample size" to "No. of Stakeholders".

Suggestion #42: Tables 5 and 6 could be combined. This will help reduce the number of tables/figures the manuscript has.
Response to suggestion #42: Table 5– Co-Design Approaches and Definition, has been removed from the manuscript. The co-design approach of each study is now included in Table 3-Summary Characteristics. Table 6-Iterative Development compared against Co-Design terminology, has been removed from the manuscript. The number of phases of each study is now included in Table 3-Summary Characteristics. We did not draw insights from comparing iterative development against co-design terminology, and as our review now focuses on learning outcomes, we have removed this table.

Suggestion #43: Tables 9 and 10 could be also combined, by adding one more column on the right in table 10 that includes the "total studies n=25" and the percentages. The references could be added as hyperindex, next to the corresponding number, similarly to tables 12 and 15.
Response to suggestion #43: Table 9 was initially used to summarize the number of studies that utilized specific involvement processes. We are now using this table to cross-classify the learning levels with the specific involvement processes. This was important for us to distinguish the specific processes that were used more frequently in the studies which were classified with higher levels of learning. Table 10 has been removed from the manuscript. We are no longer trying to extract insight from crossclassifying Co-Design Approaches with Co-Design Involvement Processes.

Suggestion #44: Tables 11 and 12 could be combined, by adding one more line at the bottom of table 12 with the total number of studies.
Response to suggestion #44: Thank you for this suggestion. Based on this recommendation and the recommendation of reviewer #1, we have moved the definitions from Table 11 to earlier in the manuscript to align with the introduction of levels of participation. Table 11 has been removed from the manuscript. We also moved Table 12 to a supplementary table, as we are not drawing extensive insight from this table, yet still wanted to provide this information to our readers.

Suggestion #45: Tables 14 and 15 could be combined, by adding one more line at the bottom of table 15, including the total number of studies
Response to suggestion #45: We have redesigned this review and no longer categorizing older adults' needs, therefore tables 14 and 15 have been removed.

Suggestion #46: Table 16 has an extra line that should be deleted, right before non-validated objective measures. It confuses the reader when trying to understand the classification presented in the table. Also, "Cognitive process" should be not bolded.
Response to suggestion #46: We have removed the extra line and adjusted cognitive processes so that it is no longer bolded.

Overall Suggestions- In general, the number of tables is quite high. As per BMJ Open's submission guidelines, the recommended number of figure and tables is "up to five", although "this is flexible, but exceeding this will impact upon the paper's 'readability'." Even if the journal allows more than 5 tables and figures, this manuscript presents 16 tables and 3 figures and most of them could be deleted or combined. While some of the tables are necessary (e.g., tables 2 and 4), others could be included as supplementary material (e.g., tables 1 or 3). Additionally, tables 5-9, 11, 13, 14 and 16 are describing each one of the review results individually (co-design approaches, iterative development, theoretical framework, study participants, co-design involvement processes, levels of participation, types of EHTs, older adults' needs, and subjective and objective measures of user testing) and some of these could be combined. Typically, in a systematic review, there are two key tables that are similar in structure: the study characteristics table and the results/outcomes table. Both of these have all the studies in the first column, one per row, and the other columns include the different variables observed for study characteristics, and results/outcomes, respectively. Although the individual tables provide important information regarding the frequency of articles that report each variable and this is helpful towards writing the results section, it makes most sense to present the majority of this information in one table (when possible), and keep the individual tables as supplementary material too. For example, the authors could add a results table including the main categories presented in table 3 (co-design approaches, stakeholder involvement and electronic healthcare tools). Lastly, in regards to tables 12 and 15 that cross-tabulate two variables, authors should consider whether the information presented is relevant and worth-mention. What can we infer from these tables? Is the information useful? Does it help to make any associations relevant for the review? Although these tables could elucidate and portray key information and associations between variables, if the answer the previous questions is no, then it is not necessary to include the tables in the manuscript.

Overall Response:
Thank you for taking the time to thoroughly review our manuscript and provide these valuable recommendations. Based on your recommendations and those of Reviewer #1, we have made significant revisions to our tables (see below).

Original Table Number	New Table Number	Table Revisions
Table 1 - Search Results	Supplementary Appendix A – Search Results	This table has been changed to a supplementary table.
Table 2 - Inclusion and Exclusion Criteria	Table 1 - Inclusion and Exclusion Criteria	No changes

Table 3 - Study Characteristics	Supplementary Appendix B- Study Characteristics	Adjusted the name of one variable, from "Sample size" to "Number of Stakeholders". This table has been changed to a supplementary table.
Table 4- Summary Characteristics	Table 3 - Summary Characteristics	This table now includes columns for Co-design Approach, Co-design Approach Defined, and No. of Phases. This allowed us to eliminate Table 5 (Co-Design Approaches and Definitions) and Table 6 (Iterative Development compared against CoDesign terminology). We have also redesigned the objective of this review and focused on learning outcomes, therefore we added two columns, Learning Level and Learning Beneficiary. For clarity, we adjusted "sample size" to "No. of Stakeholders".
Table 5 – Co-Design Approaches and Definition	Removed Table	This table has been removed from the manuscript. The co-design approach of each study is now included in Table 3.

Original Table Number	**New Table Number**	**Table Revisions**
Table 6 – Iterative Development compared against Co-Design terminology	Removed Table	This table has been removed from the manuscript. The number of phases of each study is now included in Table 3. We did not draw insights from comparing iterative development against codesign terminology, and as our review now focuses on learning outcomes, we have removed this table.
Table 7 – Theoretical framework and Co-Design Principles	Removed Table	This table has been removed from the manuscript. We have included a supplementary table that displays which frameworks were utilized by each study.

Table 8 – Study Participants	Removed Table	This table has been removed from the manuscript. This information already existed in the Summary Characteristics table, which is now Table 3.
Table 9 – Co-Design Involvement Processes	Table 6 - Co-Design Involvement Processes by Learning Levels	This table was initially used to summarize the number of studies that utilized specific involvement processes. We are now using this table to cross-classify the learning levels with the specific involvement processes. This was important for us to distinguish the specific processes that were used more frequently in the studies which were classified with higher levels of learning.
Table 10 – Co-Design Approaches cross-classified with Co-Design Involvement Processes (color-coded to reflect association in Figure 3)	Removed Table	This table has been removed from the manuscript. We are no longer trying to extract insight from this cross-classification.
Original Table Number	**New Table Number**	**Table Revisions**
Table 11 - Levels of Participation	Table 2 - Levels of Participation	We introduce this table earlier in the manuscript to provide Vaughn's definitions of the 5 levels of involvement. We removed the "Total studies" column, as this did not provide value.
Table 12 - Cross-classifying codesign approaches with levels of participation	Supplementary Appendix D - Cross-classifying co-design approaches with levels of participation	This table has been changed to a supplementary table.
Table 13 –WHO Classification of Digital Health Interventions	Table 4 –WHO Classification of Digital Health Interventions	No changes
Table 14-Older adults' needs addressed by electronic healthcare tools	Removed Table	This table has been removed from the manuscript. We have redesigned the review and are no longer categorizing older adults' needs.

Table 15 - Cross-classification of older adults' healthcare technology needs with the WHO classification of digital health interventions	Removed Table	This table has been removed from the manuscript. We have redesigned the review, and are no longer categorizing older adults' needs, therefore there is no need to cross-classify with the classification of EHTs.
Table 16 – User Testing – Subjective and Objective Measures	Table 5 – User Testing – Subjective and Objective Measures	No changes were made to the information presented; however, adjustments were made to formatting and typographical errors.
	Supplementary Appendix C	We established a results table for theoretical frameworks, specific involvement processes, user testing (endpoints, measures and validation)

We have also compared the similarities between our and Fisher's review and substantiated the differences to highlight where our review provides additional value to the readership of BMJ Open. By drawing these comparisons, it also gave us the opportunity to further reflect on additional insights we had gathered from our included studies, specifically related to learning outcomes. (see below). Learning outcomes are threaded throughout the manuscript to illustrate their relationship with the original extracted variables and are now the focal point of the review. We hope that the reviewer finds this new focus of our manuscript value-added. As a result, we were able to reduce the overall number of tables by combining relevant tables and eliminating those in which we were not drawing insight. Additional results that were extracted or summarized have been included as supplementary materials.

	Fisher et al.	**Cole et al.**
Published between	2014-2018	2010-March 2021
Main Consequences	Learning, Adjusted Design, Increased sense of participation	Usability, Errors, Performance, Need for Assistance, Meaningful experience-mutual learning (e.g., who is learning?)
User Involvement	Structured process whose outcomes are contingent on a range of premises.	User involvement is focused on whether the tool is improved.

	Synthesized results. Develop concept of user involvement. Present analytical framework.	Synthesized results. Identify gaps in mutual learning. Propose need for improving co-design frameworks.
Main Questions	1. What purposes are described in the literature for involving older people in the design of a technology? 2. How are older people selected for involvement in design practice? 3. What roles are assigned to older people when they are involved? 4. **At what design stages are older people usually involved?** 5. **At what levels of involvement are older people usually involved?** 6. What images of older people prevail in the literature involving older people? 7. What consequences of involving older users are re-ported in the literature?	1. What approaches are used for involving older adults in developing EHTs? 2. How are these approaches defined? 3. What frameworks were utilized? 4. Was iterative development utilized? How many phases? 5. **What stages of development are the approaches employed?** 6. **What is the level of involvement for older adults?** 7. What involvement processes are employed? 8. What are the subjective and objective measures used? 9. Are the subjective and objective measures validated? -------- REVISIONS NOW INCLUDE 10. Did participants reflect on testing

	Fisher et al.	**Cole et al.**
		measures? (ie., why did they make errors?) 11. Was there a bi-directional flow of knowledge between participants and researchers?
Levels of involvement	Low, Intermediate, High Arnstein (1969)	Inform, Consult, Involve, Collaborate, Empower Vaughn (2020)
Search Strategy	broad terms (a) “user involvement” and (b) “older adults.”	"co-design", "aged", "**electronic healthcare tools**",
Databases	Scopus, Web of Science	PubMed, Embase, Scopus

Analyzed	• Thematic Analysis, Coding • Differently reported consequences of • User involvement • Purposes • Stages and levels of involvement • Roles and images of older peoples • Selection procedures	• Synthesized results • How are approaches used? • How approaches are classified? • Terminologies • Frameworks • Stages of Development • Involvement processes • Levels of Participation • User testing measures ---- REVISIONS NOW INCLUDE • Learning Levels • Learning Beneficiary (Who is Learning)
User Involvement of Older People—How Does It Matter?	1. Learning, 2. Adjusted Design, and 3. Improved sense of participation. We also found consequences of older user involvement for which the literature is ambiguous	1. Learning - Bidirectional knowledge 2. Improvement in participation 3. Improved engagement in healthcare decisions
	First, from the empirical studies we reviewed, the effect of involving older users on technology ***acceptance*** remains unclear. **Second**, we found that the literature is not clear about the impact of older user involvement on technology ***adoption***.	Studies that empowered participants were more likely to utilize specific involvement processes such as contextual inquiry and functional prototyping. **Learning outcomes: *Mutual learning*** occurs when participants, designers and researchers learn from each other, or when participants learn from **reflecting** on their own efforts and errors during user testing. Learning can also occur when participants

	Fisher et al.	**Cole et al.**
		are **taught or exposed to technology prior to user testing** .
	User involvement can have different outcomes, and the type of outcomes achieved appears to differ with respect to the way designers choose to involve older people.	Learning outcomes may vary based on the types of involvement processes.

	Encourage further research on the relationship with acceptance and adoption, as well as on the actual practices of user involvement.	To ensure studies are optimizing the involvement of older adults when developing EHTs, improving existing codesign frameworks to meet the needs of older adults should be evaluated.

References

- Reference 6 and 43 are not properly referenced; they all should include the website name and the date it was accessed.
- Reference 10 is missing information. If it's an article, it is missing the journal name, volume, pages and year of publication
- In reference 19, a website link could be added
- References 23 and 25 should include the date it was accessed, not cited.

Response to reference corrections

Thank you for catching the errors in our references. All references have been updated to reflect the above suggested changes.

VERSION 2 – REVIEW

REVIEWER	de Mul, Marleen Erasmus Universiteit Rotterdam Erasmus School of Health Policy and Management
REVIEW RETURNED	15-Apr-2022

GENERAL COMMENTS	Thank you for your extensive reponse to my previous comments. The paper has much improved, because of the new focus. I accept this paper, provided the following points are addressed: Abstract I am not sure what is meant with the last sentence of the abstract Strengths and limitations EHT is already mentioned in the first bullet, so abbreviation should go there as well. I do not really see a difference between bullet 1 and bullet 4, so I do not see why bullet 4 is a limitation. It could be a limitation if there are specific characteristics to be considered a EHT for this study, though. Aim All but one aim are in past tense Results Table 3: Personally, I would leave out the column Diagnosis (which is for me not relevant for this study) and add as column 5 the Digital Health Intervention, using only the six main categories. Or describing the intervention in normal words (website, app, telemonitoring device, etc.) Maybe study setting is also not that relevant, it is not referred to in the actual analysis (only description of characteristics) Types of Electronic Healthcare Tools (EHTs): Is there redundancy in sentence 2 and 3?

	Table 6: is there an alternative for “contextual inquiry” in the column “involvement process”? it lack information, since it is the same as the first column. I would imagine that in the contextual inquiry phase also interviews or surveys are done? Where can I find information about the participation level (without mapping to co-design approach) per study? Figure 3 is difficult to read in this format with cumulative percentages. For example, I do not know from how many cases each bar is constructed, and what the Yes and No bar show. Also there seems to be an error in the final 2 columns (should be Consult (No)). I feel having one bar per phase (instead of per participation level) is more clear, demonstrating which level of participation is achieved per phase. That makes it easier to conclude about where in the whole design processes the older adults are used most as partners. Discussion Key findings: is there any conclusion about the types of EHT developed and the way older adults are involved? Co-design approaches: An interesting point for the discussion is whether there are studies that are presented as codesign, which they in fact are not, because there is no mutual exchange, that they are only involved as interviewee for example, but not as a participant. I think here you can be more critical. Electronic Health Tools: could it be that the user experiences are reported in a subsequent paper by the research group? That would explain little information about the user testing in the design paper? I am sure there are many feasibility and usability studies in this field, which would not be in your sample. Conclusion: Last sentence is not clear. Do you mean that frameworks should be improved or that better frameworks should be applied? Supplementary material Supplementary appendix C: please add the reference number to the author column I look forward to future work by your research group in this area. It is definitely important that codesign methodology for older adults is further improved.

REVIEWER	Otero Varela, Lucía University of Calgary, Community Health Sciences
REVIEW RETURNED	08-May-2022

GENERAL COMMENTS	Thank you for your resubmission, which demonstrates significant efforts have been made to improve the paper by taking on board the comments made during the initial review process. The subsequent paper is now more robust, readable, and academically proficient. A couple of further minor recommendations, ordered by manuscript section are below: Introduction - Page 7, line 34-48. The aim of the study should include the primary and secondary purposes of the review, for instance in the form of research questions. The current questions simply constitute a list of the study data that will be extracted from the articles, which is already explained in the methods section. It should not be included as part of the aims.

	Methods - Page 9, line 3. Supplementary Appendix B presents useful information to better understand how study characteristics were obtained (e.g., metadata, references for definitions, …). However, the name for this appendix (Study Characteristics) is incorrect, as that would imply the actual studies and their characteristics, like Table 3 (summary characteristics) does. - Page 9, line 27. Table 4 should not be mentioned in the methods, as it is a table with results. - Page 12, line 6-7. “Supplementary Appendix C includes the full list of involvement processes used within each study.” Should be part of the results, not the methods section. Results - Page 10, line 42. The “Results of Systematic Review” section name is not necessary, the authors can start directly with the paragraph “The initial database search returned 835 articles…”.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Dr. Marleen de Mul, Erasmus Universiteit Rotterdam Erasmus School of Health Policy and Management

Comments to the Author:
Thank you for your extensive reponse to my previous comments. The paper has much improved, because of the new focus. I accept this paper, provided the following points are addressed:
I look forward to future work by your research group in this area. It is definitely important that codesign methodology for older adults is further improved.

Overall Response to Reviewer 1:

Thank you for your insightful comments and interest in our future research. We appreciate the time and effort you have dedicated in providing valuable feedback on our manuscript. We believe the suggested improvements will make this manuscript more valuable to the readers of BMJ Open.

Suggestion 1: Abstract
I am not sure what is meant with the last sentence of the abstract

Response to suggestion 1:

Thank you for bringing this to our attention, we have clarified the last sentence of the abstract.

Change made:

To ensure studies optimize participation and learning of older adults when developing EHTs, there is a need to place more emphasis on the approaches promoting mutual learning.

Suggestion 2: Strengths and limitations

EHT is already mentioned in the first bullet, so abbreviation should go there as well.
I do not really see a difference between bullet 1 and bullet 4, so I do not see why bullet 4 is a limitation. It could be a limitation if there are specific characteristics to be considered a EHT for this study, though.

Response to suggestion 2:

Thank you for observing the redundancy of bullet points #1 and #4, we have chosen to delete point #4. We have also included the abbreviation (EHT) in bullet point #1.

Changes made:

- This review provides a comprehensive assessment of the utilization of co-design approaches in studies that involve older adults in developing electronic healthcare tools EHTs.
- We developed a broad search strategy to ensure all co-design or “collaborative” approaches were captured in this review.
- We utilized widely reported classification methods to categorize the included studies.
- We limited inclusion criteria to studies that involved older adults in the development of an electronic healthcare tool (EHT).
- We limited our analysis to the overall involvement processes and levels of participation at the study level; cross-classification of these variables at the design stage level for each study was not conducted.
- This review was limited to initial studies and did not evaluate whether subsequent papers by research groups provided a broader analysis of user experience.

Suggestion 3: Aim
All but one aim are in past tense

Response to suggestion 3:

Thank you for noticing this discrepancy. Our aims are now consistently presented in the past tense.

Changes made:

- What approaches were used for involving older adults in developing EHTs and how were they defined?
- What theoretical frameworks and design principles were used to develop EHTs for older adults?
- Was iterative development utilized? If so, how many phases were conducted?
- What were the subjective and objective measures used for study endpoints, and were the measures validated?
- What involvement processes were employed?
- What were the levels of participation for older adults?
- Did participants reflect on testing measures? (i.e., reflect on why an error was made)

- Was there a bi-directional flow of knowledge between participants and researchers?

Results

Suggestion 4: Table 3: Personally, I would leave out the column Diagnosis (which is for me not relevant for this study) and add as column 5 the Digital Health Intervention, using only the six main categories. Or describing the intervention in normal words (website, app, telemonitoring device, etc.) Maybe study setting is also not that relevant, it is not referred to in the actual analysis (only description of characteristics)

Response to suggestion 4:

Thank you for providing this perspective. We have evaluated both the diagnosis and study setting fields and have chosen to replace the study setting with your suggested column for summarizing Digital Health Interventions. We looked at categorizing each tool using normal words (website, app, telemonitoring device, etc.), but preferred using The World Health Organization's 6 categories of Digital Health Interventions, as this was consistent with how we defined each EHT. After reviewing and analyzing studies further, as you recommended in suggestion #8, we felt confident making the decision to use the WHO's categorization over normal words.

Changes made:

First Author's last name	Year of Pub	First author's location	*Digital Health Intervention	Co-Design Approach
Abujarad [1]	2021	USA	PHT CBR ODISC	User Centered Design
Ahmed [2]	2019	USA	TCC PHT	Participatory Design

*TCC = Targeted Client Communication, CCC = Client to Client Communication, PHT = Personal Health Tracking, CBR = Citizen Based Reporting, ODIFC = On-Demand Information Services to Clients

Suggestion 5: Types of Electronic Healthcare Tools (EHTs): Is there redundancy in sentence 2 and 3?

Response to suggestion 5:

Thank you for catching this redundancy. We have deleted sentence 3 from this section.

Changes made:

Table 4 summarizes the EHTs based on the WHO Classification of Digital Health Interventions. In 12 of the 25 studies, the EHT that was developed or redesigned fits within multiple categories, therefore, was cataloged in multiple categories. Of the 25 studies, 12 developed a tool that spanned multiple categories. The overall objective was to ensure the tools included in the selected studies fit within a categorization that could be defined as those which support healthcare decisions, which was acieved through this classification process.

Suggestion 6: Table 6: is there an alternative for "contextual inquiry" in the column "involvement process"? it lacks information, since it is the same as the first column. I would imagine that in the contextual inquiry phase also interviews or surveys are done?

Response to suggestion 6:

Thank you for the suggestion. Leinonen et al. (2008) proposed research-based design that includes 1) contextual inquiry, 2) participatory design, 3) product design and 4) software as hypothesis, a four-phased iterative process. In the contextual inquiry phase, the goal is to understand the context and the preliminary design challenges. It involves understanding the user, what the design challenges are and why they are being solved. Leinonen et al. proposed ethnographic methods and benchmarking the environment to have a complete understanding of the major design challenges. Therefore, we did not change the name of the first phase "contextual inquiry." The major ethnographic method that we noticed in this phase is "contextual inquiry" and hence there is a redundancy in the column "involvement processes" and the "phase." Our preference is to use Leinonen et al.'s terminology but to avoid confusion for the readers, we have replaced contextual inquiry with "rapid ethnography" as reported by Durall and Leinonen et al (2021) [1] in a recent study.

Changes made:

Table 6 – Co-Design Involvement Processes by Learning Levels

(color-coded to reflect association in Figure 3)

	Involvement Processes	***Level 0 (n=25), n(%)***	***Single Loop Level 1 (n=8), n(%)***	***Double Loop Level 2 (n=7), n(%)***
***Contextual Inquiry** (Rapid Ethnography)	Shadowing in context including semi/unstructured interviews, observation, etc.	6 (24%)	4 (50%)	4 (57%)
Participatory Design	Survey	10 (40%)	3 (38%)	3 (43%)
	Interview	18 (72%)	7 (88%)	5 (71%)
	Cognitive walkthrough	8 (32%)	4 (50%)	3 (43%)
	Think aloud	11 (44%)	2 (25%)	1 (14%)

	Remote evaluation	1 (4%)	1 (13%)	1 (14%)
	Focus group	8 (32%)	4 (50%)	4 (57%)
	Task analysis	10 (40%)	3 (38%)	2 (29%)
	Scenarios	7 (28%)	2 (25%)	1 (14%)
	Prototype (low fidelity)	8 (32%)	4 (50%)	3 (43%)
	Working Group Workshop	9 (36%)	5 (63%)	3 (43%)
Product Design	Use Cases	4 (16%)	2 (25%)	1 (14%)
	User Stories	9 (36%)	2 (25%)	2 (29%)
	Prototype (medium fidelity/throw-away)	18 (72%)	8 (100%)	5 (71%)
	Heuristic evaluation	5 (20%)	2 (25%)	1 (14%)
Software Prototype as Hypothesis	Prototype (functional/Beta version)	15 (60%)	8 (100%)	7 (100%)
Other	Other	5 (16%)	3 (38%)	2 (29%)
Total		**152**	**64**	**48**

*The goal of contextual inquiry is to understand the context and the preliminary design challenges. It involves understanding the user, what the design challenges are and why they are being solved. Leinonen et al. [24] proposed ethnographic methods and benchmarking the environment to have a complete understanding of the major design challenges. Contextual inquiry is used as a category to remain consistent with Leinonen et al.'s terminology, but we recognize that this category can be broadened into "rapid ethnography" which Leinonen recommended in his follow-up research contributions [56].

Suggestion 7: Where can I find information about the participation level (without mapping to co-design approach) per study? Figure 3 is difficult to read in this format with cumulative percentages. For example, I do not know from how many cases each bar is constructed, and what the Yes and No bar show. Also there seems to be an error in the final 2 columns (should be Consult (No)). I feel having one bar per phase (instead of per participation level) is clearer, demonstrating which level of participation is achieved per phase. That makes it easier to conclude about where in the whole design processes the older adults are used most as partners.

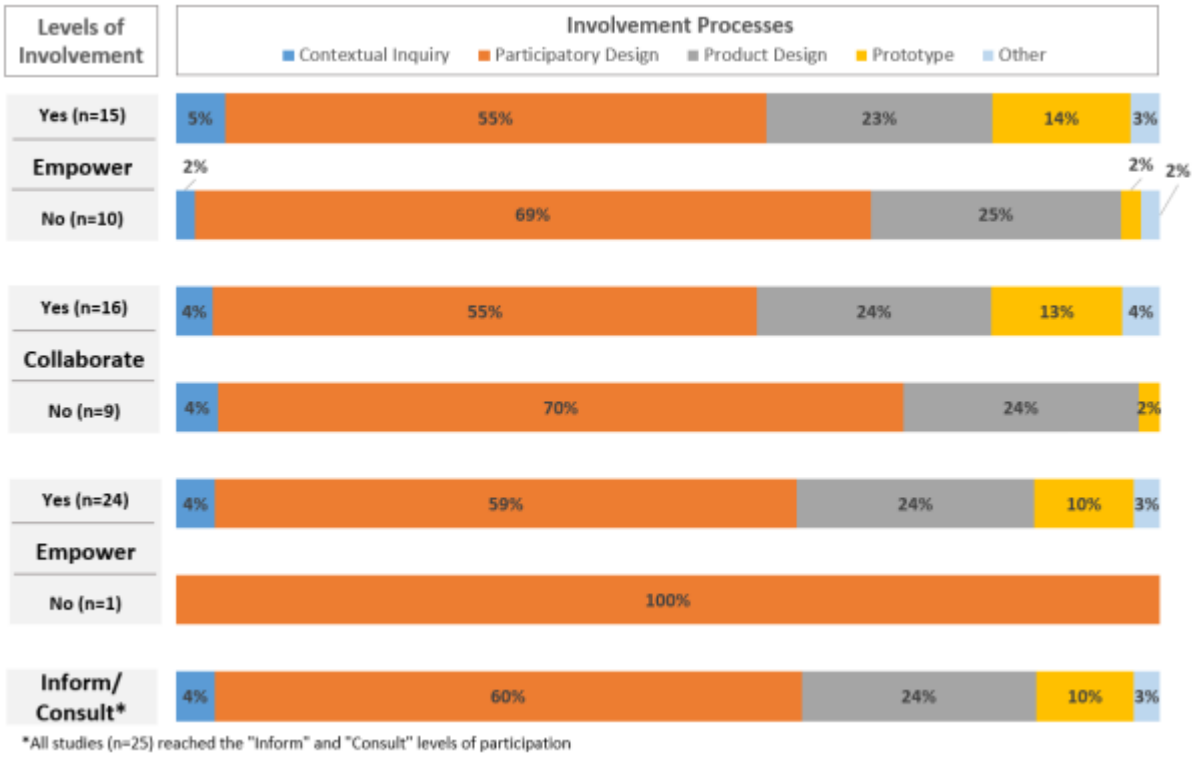
Response to suggestion 7:

Thank you for making this point. We have revised Supplementary Appendix C to include “Levels of Participation.” Appendix C also includes all involvement processes conducted within each study. This additional field will now allow readers to map involvement processes to the levels of involvement. We have reviewed our options for breaking this down by phase, but believe this would be an imprecise analysis, as studies did not consistently specify which processes were conducted within specific phases. We have also revised Figure 3 to consolidate the “inform” and “consult” levels. All studies met these levels; therefore, we do not have a “no” row for either level. We also created an alternate figure, (see below) and would appreciate guidance as to which figure conveys the cross mappings most effectively.

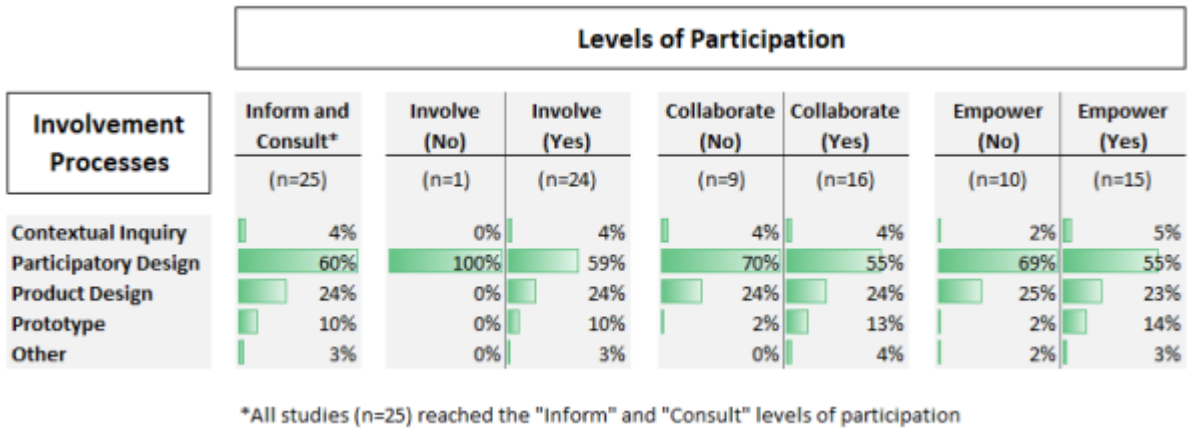
Figure 3 – Original:



Changes made to Figure 3:



Alternate for Figure 3:



Discussion

Suggestion 8: Key findings: is there any conclusion about the types of EHT developed and the way older adults are involved?

Response to suggestion 8:

Following suggestion #4, to either add a column for digital health intervention or describe this intervention in normal words (website, app, telemonitoring device, etc.), we evaluated each study to determine whether there were differences in the way older adults were involved when comparing the type of EHT developed. First, we designated each study using normal words and compared against “levels of involvement”, “types of involvement processes”, and “learning levels”. We then designated

each study by the six main categorizations defined by the World Health Organization and compared against the same variables. These analyses did not result in any key findings to suggest a difference in the way older adults were involved when comparing with the EHT developed.

Suggestion 9: Co-design approaches: An interesting point for the discussion is whether there are studies that are presented as codesign, which they in fact are not, because there is no mutual exchange, that they are only involved as interviewee for example, but not as a participant. I think here you can be more critical.

Response to suggestion 9:

Thank you for suggesting we take the opportunity to be more critical regarding studies that present themselves as using a co-design approach, but only engage participants to receive feedback, rather than to engage them in mutual learning. Based on your feedback, we have made the following revisions to our discussion section > co-design approaches:

Changes made:

However, studies did not specify the extent to which older adults should be involved, nor did they specify whether a mutual learning model should be employed. Of interest, is that only 36% (n=9) of included studies had a mutual exchange of knowledge between the research team and study participants, even though all studies were classified as having utilized a co-design approach. For the remaining studies (n=16; 64%), participants engaged in processes in which they simply provided feedback on the EHT and did not engage in processes that facilitated a mutual exchange of knowledge. Future co-design studies involving older adults should utilize improved co-design frameworks and optimize mutual learning.

Suggestion 10: Electronic Health Tools: could it be that the user experiences are reported in a subsequent paper by the research group? That would explain little information about the user testing in the design paper? I am sure there are many feasibility and usability studies in this field, which would not be in your sample.

Response to suggestion 10:

Thank you for this clarification. We agree that it is possible that user experience may be reported in subsequent papers and possibly explain for limited information about user testing in the included studies for this review. We have included this suggestion in both the “Strengths and Limitations” of this Study”, as well as in the discussion section, under limitations.

Changes made:
Strengths and Limitations of this study

- This review provided a comprehensive assessment of the utilization of co-design approaches in studies that involve older adults in developing electronic healthcare tools (EHTs).
- We developed a broad search strategy to ensure all co-design or “collaborative” approaches were captured in this review.
- We utilized widely reported classification methods to categorize the included studies.
- We limited our analysis to the overall involvement processes and levels of participation at the study level; cross-classification of these variables at the design stage level for each study was not conducted.

- This review was limited to initial studies and did not evaluate whether subsequent papers by research groups provided a broader analysis of user experience.

Discussion > Limitations

As our inclusion criteria spanned an array of terms that we categorized as co-design approaches, we had several approaches that were only represented by one study, which may not fully represent all studies that incorporate such approaches. One of the biggest limitations of this study was the difficulty in extracting and classifying the information, however we used mitigation strategies (e.g., established frameworks and classification methods) to support the review of variables including types of EHTs, stakeholder involvement and participation and levels of learning. Studies that employed co-design approaches were included in our review, however, we limited our review to the research groups' initial study and did not evaluate whether subsequent papers by these groups provided a broader analysis of user experience.

Suggestion 11: Conclusion: Last sentence is not clear. Do you mean that frameworks should be improved or that better frameworks should be applied?

Response to suggestion 11:

Thank you for this suggestion. We have clarified our original sentence "To ensure studies are optimizing the involvement processes, participation and learning of older adults when developing EHTs, improving existing co-design frameworks to meet the needs of older adults should be evaluated." and made the following change.

Change made:

To ensure studies are optimizing the involvement processes, participation and learning of older adults when developing EHTs, there is a need to place more emphasis on the approaches promoting mutual learning.

Supplementary material
Suggestion 12: Supplementary appendix C: please add the reference number to the author column

Response to suggestion 12:

References numbers have been added to the author column in supplementary appendix C.

Change made:

Author	**Theoretical Framework/Co-Design Principle**	**Level of Involvement**
Abujarad [52]	User Centered Design Approach	Empower
Ahmed	Participatory Design Approach	Empower

[34]	Developed recommendations for framework	
Bogza [29]	Center for eHealth and Wellbeing Research guidelines Inductive thematic qualitative analysis approach	Empower

Reviewer: 2
Dr. Lucía Otero Varela, University of Calgary

Comments to the Author:
Thank you for your resubmission, which demonstrates significant efforts have been made to improve the paper by taking on board the comments made during the initial review process. The subsequent paper is now more robust, readable, and academically proficient. A couple of further minor recommendations, ordered by manuscript section are below:

Overall Response to Reviewer 2:

Thank you for your insightful comments. We appreciate the time and effort you have dedicated in providing valuable feedback on our manuscript. We believe the incorporated changes that reflect your suggestions are value added for the readers of BMJ Open.

Introduction
Suggestion 1: Page 7, line 34-48. The aim of the study should include the primary and secondary purposes of the review, for instance in the form of research questions. The current questions simply constitute a list of the study data that will be extracted from the articles, which is already explained in the methods section. It should not be included as part of the aims.

Response to suggestion 1:

Thank you for your suggestion. You have raised an important point here. However, we believe that our research questions drive the data that was extracted from the studies included in the review and not the vice versa. This approach has also been utilized in other peer reviewed systematic reviews [2]. To improve this further, based on your suggestion, we have reorganized our questions as primary and secondary research questions:

Change made:

Primary research question:

This systematic review Synthesizes the current state of utilizing co-design approaches with older adults in the development of electronic healthcare tools. and seeks to answer the following questions.

Secondary research questions:

- What approaches were used for involving older adults in developing EHTs and how were they defined?
- What theoretical frameworks and design principles were used to develop EHTs for older adults?
- Was iterative development utilized? If so, how many phases were conducted?
- What were the subjective and objective measures used for study endpoints, and were the measures validated?
- What involvement processes were employed?
- What was the level of participation for older adults?
- Did participants reflect on testing measures? (i.e., reflect on why an error was made)
- Was there a bi-directional flow of knowledge between participants and researchers?

Methods

Suggestion 2: Page 9, line 3. Supplementary Appendix B presents useful information to better understand how study characteristics were obtained (e.g., metadata, references for definitions, …). However, the name for this appendix (Study Characteristics) is incorrect, as that would imply the actual studies and their characteristics, like Table 3 (summary characteristics) does.

Response to suggestion 2:

Thank you for suggesting we revised the name of this supplementary table. We have now titled this, “Study Classification Methods”.

Change made:

Supplementary Appendix B - Study Classification Methods

Category	***Variable***	***Classification method***
Summary characteristics	Year of publication	Metadata
	First author’s geolocation	Metadata
	Study setting	Metadata
	Number of stakeholders	Metadata, mean
	Number of older adults	Metadata, mean
	Study population	Metadata

Suggestion 3: Page 9, line 27. Table 4 should not be mentioned in the methods, as it is a table with results.

Response to suggestion 3:

Thank you for the suggestion. We agree with this and have deleted the reference to Table 4 in methods.

Change made: The classification tool for clients consists of 7 categories, each further broken down into subcategories, as depicted in Table 4

Suggestion 4: Page 12, line 6-7. "Supplementary Appendix C includes the full list of involvement processes used within each study." Should be part of the results, not the methods section.

Response to suggestion 4:

Thank you for the suggestion. We agree with this and have made the following revisions.

Change made: Moved the following sentence from Methods > Data Extraction and Synthesis to Results >Types of Involvement Processes.

A variety of involvement processes were used in the 25 studies, as indicated in Table 6, which reports 152 processes across all the included studies. Supplementary Appendix C includes the full list of involvement processes used within each study. The involvement processes…

Results

Suggestion 5: Page 10, line 42. The "Results of Systematic Review" section name is not necessary, the authors can start directly with the paragraph "The initial database search returned 835 articles…".

Response to suggestion 5:

We have removed the header, "Results of Systematic Review".

Change made:

Results

Results of Systematic Review

The initial database search returned 835 articles, including 495 from PubMed, 246 from Embase, and 94 from Scopus, of which 210 were duplicates, leaving 625 articles for the title and abstract review.

We again appreciate the opportunity to resubmit our manuscript with the above changes reflecting the reviewers' comments. On behalf of all the authors, thank you for considering our manuscript for publication in BMJ Open

Sincerely,

Amy Cole, MPS, MS

University of North Carolina, Carolina Health Informatics Program

Chapel Hill, NC 27599

References:

1 Durall E, Perry S, Hurley M, *et al*. Co-Designing for Equity in Informal Science Learning: A Proof-of-Concept Study of Design Principles. *Front Edu*c 2021**;**6. doi:10.3389/feduc.2021.675325

2 Quintero J, Baldiris S, Rubira R, *et al.* Augmented reality in educational inclusion. A systematic review on the last decade. *Front Psychol* 2019;**10**:1835. doi:10.3389/fpsyg.2019.01835

VERSION 3 – REVIEW

REVIEWER	de Mul, Marleen Erasmus Universiteit Rotterdam Erasmus School of Health Policy and Management
REVIEW RETURNED	19-Jun-2022

GENERAL COMMENTS	Thank you for making the final improvements to the paper. It was a joy to read.

REVIEWER	Otero Varela, Lucía University of Calgary, Community Health Sciences
REVIEW RETURNED	17-Jun-2022

GENERAL COMMENTS	Thank you for addressing my previous comments. The paper is now more robust, readable, and academically proficient. I recommend this paper for publication, although I would like to make one final clarification in regards to the research question. The primary research question is now worded as an aim for the review (“synthesizing the literature”). It should be worded as a question instead. For instance, following the example you provided [2], it could read: “What is the current state of utilizing co-design approaches with older adults in the development of electronic healthcare tools?”. Otherwise, it would be better to return to the original sentence.